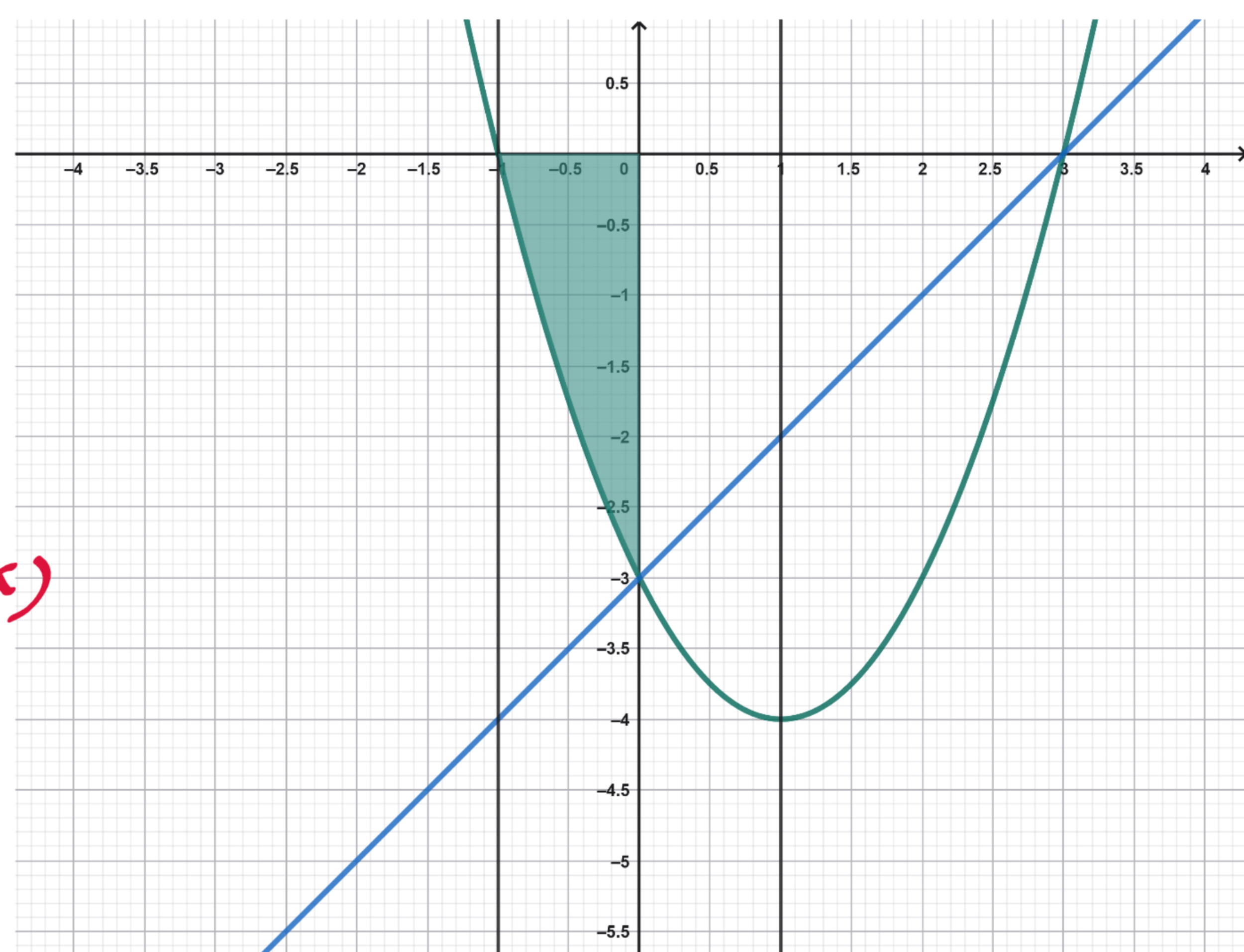
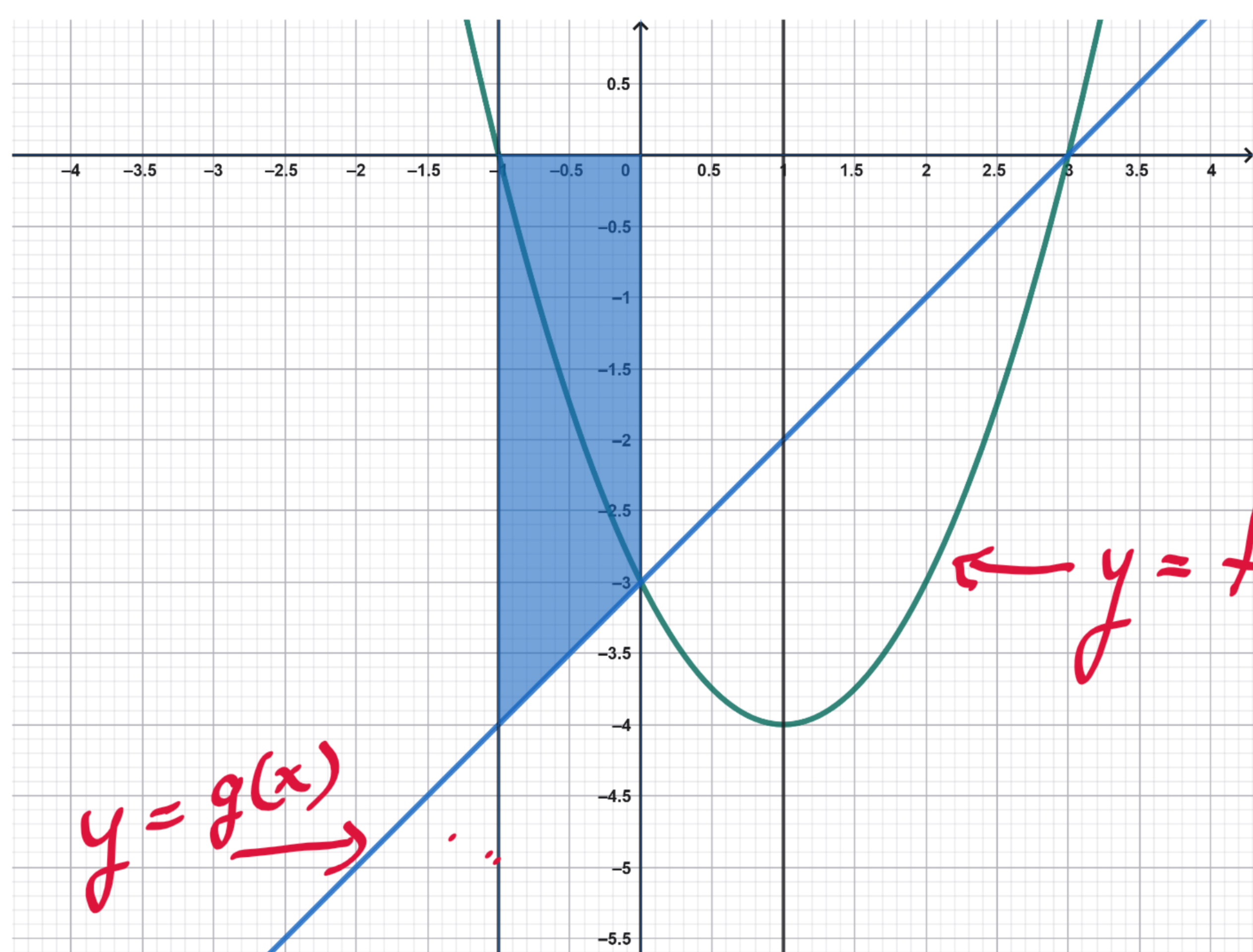




$$A_1 = -\int_{-1}^0 g(x) dx$$

$$A_2 = -\int_{-1}^0 f(x) dx$$

$$A = A_1 - A_2$$

$$= -\int_{-1}^0 g(x) dx - \left( -\int_{-1}^0 f(x) dx \right)$$

$$= -\int_{-1}^0 g(x) dx + \int_{-1}^0 f(x) dx =$$

$$= \int_{-1}^0 (f(x) - g(x)) dx$$

$$y^2 = 2x + 6$$

y-axis.

$$y^2 = 2x + 6$$

$$2x + 6 = y^2$$

$$2x = y^2 - 6$$

$$x = \frac{1}{2} y^2 - 3$$

$$(-3, 0)$$

$$x = 0$$

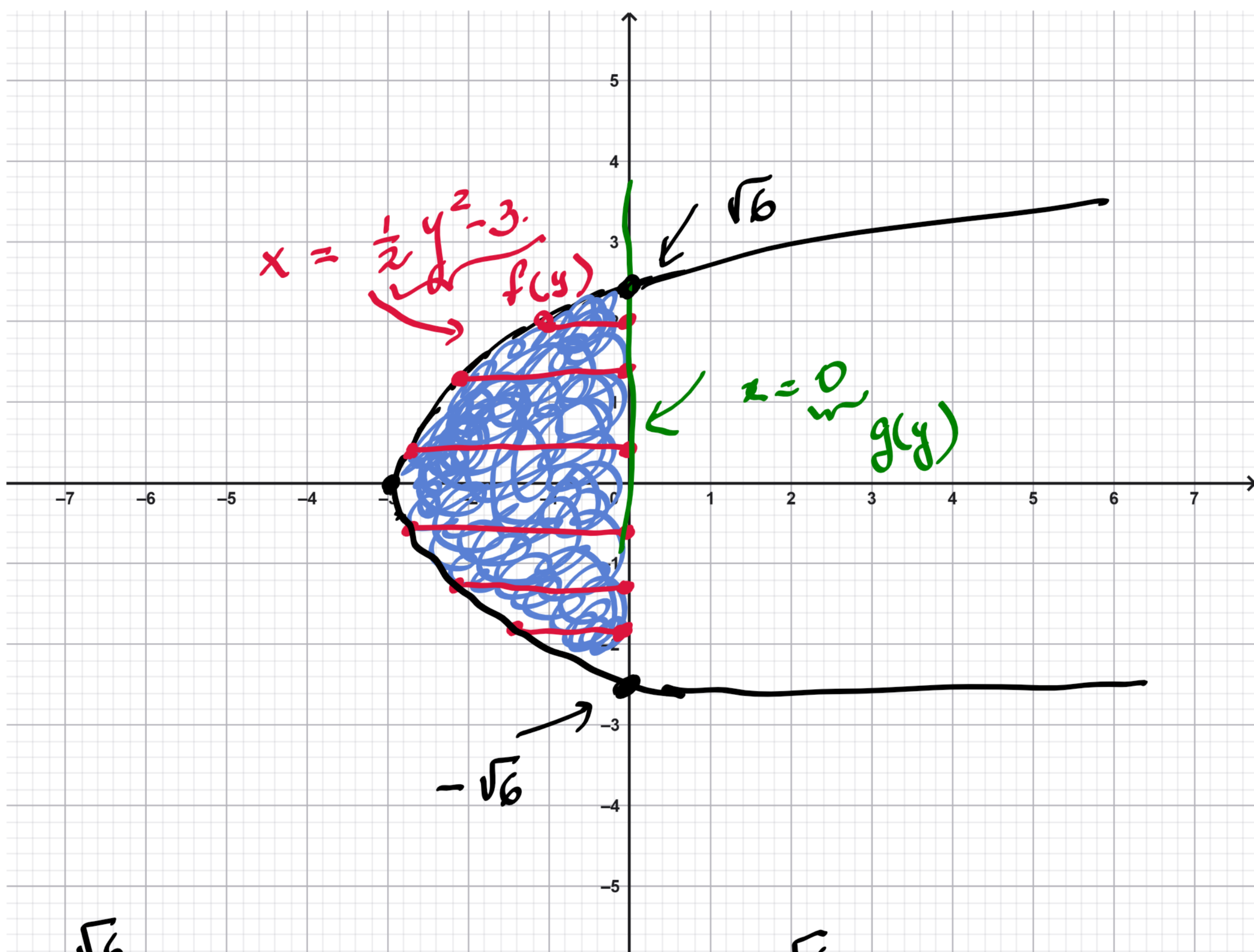
$$y^2 = 6$$

$$|y| = \sqrt{6}$$

$$y = \sqrt{6} \quad \text{or} \quad y = -\sqrt{6}$$

$$- \left( -\frac{1}{6} (-\sqrt{6})^3 \right)$$

$$\underbrace{(-1)^3 (\sqrt{6})^3}_{-(\sqrt{6})^3}$$



$$\int_{-\sqrt{6}}^{\sqrt{6}} (0 - (\frac{1}{2} y^2 - 3)) dy = \int_{-\sqrt{6}}^{\sqrt{6}} (3 - \frac{1}{2} y^2) dy$$

$$= \left( 3y - \frac{1}{6} y^3 \right) \Big|_{-\sqrt{6}}^{\sqrt{6}}$$

$$= \left( 3\sqrt{6} - \frac{1}{6} (\sqrt{6})^3 \right) - \left( -3\sqrt{6} - \frac{1}{6} (-\sqrt{6})^3 \right)$$

$$= 3\sqrt{6} - \frac{1}{6} (\sqrt{6})^3 + 3\sqrt{6} - \frac{1}{6} (\sqrt{6})^3$$

$$= 6\sqrt{6} - \frac{1}{3} (\sqrt{6})^3$$

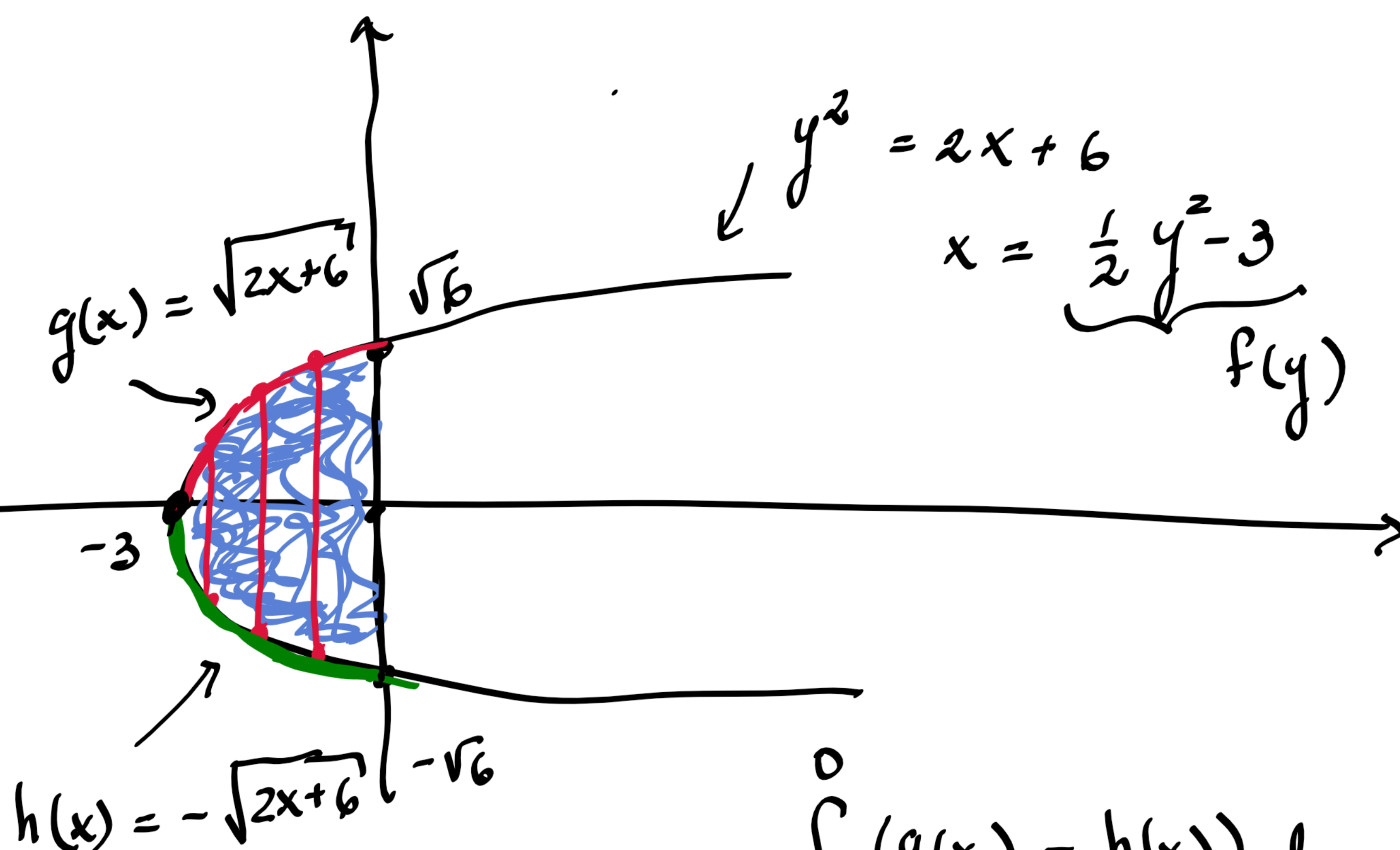
$$\underbrace{(\sqrt{6})^2}_{=6} \sqrt{6} = 6\sqrt{6}$$

$$y^2 = 2x + 6$$

$$|y| = \sqrt{2x+6}$$

$$y = \sqrt{2x+6}$$
  
or

$$y = -\sqrt{2x+6}$$



$$y^2 = 2x + 6$$
  

$$x = \frac{1}{2} y^2 - 3$$
  

$$f(y)$$

$$\int_{-3}^0 (g(x) - h(x)) dx$$
  

$$= \int_{-3}^0 2\sqrt{2x+6} dx$$